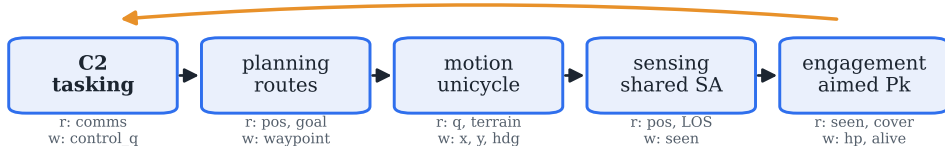


One simulation tick, and the entity state it touches

advance one tick: $t \rightarrow t + \Delta t$ ($\Delta t = 1$ s, fixed order, no per-agent FSM)



Entity state today (SoA, one row per platform)

kinematics	x, y, z, heading, speed
status	side, ptype, role, alive, hp, fuel
move target	tgt_x, tgt_y
platform	max_speed, turn_rate, sensor_range, weapon_range, pk_base, domain
perception	seen (1 bit, per tick)
control	control_quality, await_until, decision_cooldown

Proposed, AFSIM-aligned (not built)

belief / tracks (per side)

believed x,y . age (staleness) . confidence
fires resolve vs belief, not truth
false / decoy tracks -> EW, spoofing,
decoys become first-order

suppression	decaying; degrades fire/detect
munitions	by class; deplete on fire
C2 graph	command chain / connectivity
sustainment	fuel + fatigue / posture

*The fixed 1 s tick is the low-fidelity reduction of AFSIM's discrete-event executive;
the belief layer mirrors AFSIM's perception processor and false-target / track effects.*